

QUESTIONS

Multiple Choice Questions (Choose 1 out of 3)

1. How many types of flowers exist based on the presence of reproductive parts?
A) 2 types (which have only male or only female parts)
B) 3 types (only female, only male, or both parts)
C) 4 types
2. What forms the corolla of a flower?
A) The sepals
B) The anther and filament
C) All the petals
3. Which of the following act as pollinators during the pollination stage?
A) Wind, bees, and butterflies
B) Rain and snow
C) Roots and leaves
4. What is the "petiole" in a leaf?
A) The tiny openings that control gases
B) The part attached to the stem
C) The wide part of the leaf
5. During photosynthesis, what directly captures sunlight in the leaves?
A) The roots
B) The chlorophyll
C) The raw sap
6. In which stage of photosynthesis is oxygen released into the air?
A) Stage 1, when roots absorb raw sap
B) Stage 3, when stomata take in carbon dioxide
C) Stage 7, released by the stomata
7. Which of these characteristics describes the leaves of evergreen trees?
A) They are broad and turn yellow in autumn
B) They are thin, needle-like, and remain green all year
C) They lose all their leaves in winter
8. What happens to the ovules during seed formation?
A) They develop into seeds
B) They grow to form the calyx
C) They evaporate through the leaves
9. What is the process in which plants evaporate extra water through the leaves?
A) Respiration
B) Photosynthesis
C) Transpiration
10. What is the function of the veins in a leaf?
A) To attract pollinators with bright colors
B) To run through the leaf like tiny pipes, bringing water and nutrients
C) To connect the stigma to the ovary

True or False Questions

1. Small green leaves form the calyx that holds together the petals.
2. The stigma is the sticky part of the carpel which collects pollen from other flowers.
3. For seed dispersal to be successful, seeds only need fertile soil to grow into new plants.
4. The "lamina" represents the wide part of the leaf.
5. Stomata are tiny openings on the underside of the leaf which control gases going in and out.
6. Sunlight activates chlorophyll photosynthesis, meaning raw sap combines with carbon dioxide and becomes processed sap.
7. In deciduous trees, when it's cold, the amount of chlorophyll decreases and leaves change colour.
8. Evergreen trees don't lose all their leaves in winter; instead, some old leaves fall and new ones grow all year.
9. Plant respiration (absorbing oxygen and releasing carbon dioxide) happens only during the night.
10. The ovary contains ovules which develop into seeds once fertilized.

Open-Ended Questions

1. What anatomical parts form the female part of the flower (the pistil), and what does the ovary do?
2. What are the three different stages of flowering and what briefly happens in each stage?
3. Can you summarize the seven stages of the photosynthesis process?
4. What are the main differences between deciduous trees and evergreen trees based on their leaves?
5. What three elements do seeds need to grow into new plants after seed dispersal?

ANSWERS

Answers to Multiple Choice Questions

1. **B** (There are 3 types of flowers: those with only female parts, only male parts, or both parts).
2. **C** (The corolla is formed by all the petals).
3. **A** (Pollen grains are transferred by different pollinators: wind, bees, butterflies).
4. **B** (The petiole is the part attached to the stem).
5. **B** (Chlorophyll in the leaves captures sunlight).
6. **C** (In stage 7, stomata release oxygen in the air).
7. **B** (Evergreen trees have thin, needle-like leaves that remain green all year).
8. **A** (The ovules develop into seeds).
9. **C** (Transpiration is the process in which plants evaporate extra water).
10. **B** (Veins run through the leaf like tiny pipes, bringing water and nutrients).

Answers to True or False Questions

1. **True.** Small green leaves form the calyx that holds together the petals.
2. **True.** The stigma is the sticky part of the carpel which collects pollen from other flowers.
3. **False.** To grow into new plants, seeds need fertile soil, water, and sunlight.
4. **True.** The lamina is the wide part of the leaf.
5. **True.** Stomata are tiny openings on the underside which control gases going in and out.
6. **True.** Sunlight activates chlorophyll photosynthesis: raw sap combines with carbon dioxide to become processed sap.
7. **True.** When it's cold, the amount of chlorophyll decreases and leaves change colour.
8. **True.** Evergreen trees remain green all year; trees don't lose all their leaves in winter, some old leaves fall and new ones grow all year.
9. **False.** Respiration is the process in which plants absorb oxygen and release carbon dioxide, and it happens during the day and the night.
10. **True.** The ovary contains ovules which develop into seeds once fertilized.

Answers to Open-Ended Questions

1. The pistil is the female part of the flower and is formed by the stigma, the style, and the ovary. The ovary contains ovules which develop into seeds once fertilized.
2. The three stages are: *Pollination*, where pollen grains are transferred from the anther to the stigma; *Fertilisation*, where the pollen arrives at the ovary and fertilises the ovules; and *Seed Formation*, where the ovules develop into seeds and the ovary becomes the fruit.
3. The seven stages are: 1) Roots absorb raw sap from the soil; 2) Raw sap is transported to the stem and leaves; 3) Stomata take in carbon dioxide from the air; 4) Chlorophyll in the leaves captures sunlight; 5) Sunlight activates chlorophyll photosynthesis, combining raw sap with carbon dioxide to become processed sap and releasing oxygen; 6) Processed sap is transported to the other parts of the plant; 7) Stomata release oxygen in the air.
4. Deciduous trees have broad leaves that are green in summer, but turn yellow and fall in autumn. Evergreen trees have thin, needle-like leaves that remain green all year, and they don't lose all their leaves in winter.
5. After being carried far away from the parent plant during seed dispersal, seeds need fertile soil, water, and sunlight to grow into new plants.